

Percentages



1 "out of 100"

2

a 100

b 53

3

a

i 71

ii 71%

iii $\frac{71}{100}$

b

i 50

ii 50%

iii $\frac{1}{2}$

Converting from percentages

4

a $\frac{17}{25}$
e $\frac{37}{100}$

b $\frac{13}{100}$
f $\frac{47}{50}$

c $\frac{11}{20}$
g $\frac{9}{2}$

d $2\frac{3}{4}$
h $\frac{1}{40}$

5

a 0.54
e 0.76

b 0.6
f 5

c 2.7
g 1.82

d 3.03
h 68.97

Converting to percentages

6

a i $\frac{50}{100}$
ii 50%

b i $\frac{3}{100}$
ii 3%

c i $\frac{12.5}{100}$
ii 12.5%

d i $\frac{22.5}{100}$
ii 22.5%

7

a 83%
e 62%

b 180%
f 60%

c 42%
g 370%

d 65%
h 218%

8

- a No
b 67%



9

- a 13% b $\frac{4}{25} = 16\%$ c $\frac{63}{90} = 70\%$ d $\frac{1}{3}$

10

- a 24% b 110% c 85.7% d 800%
e 171% f 2400% g 1304% h 4.1%

Comparing different values

11 Multiplying by 100% introduces the % symbol without changing the number's value.

12 Dividing by 100% removes the % symbol without changing the number's value.

13

Fraction	Decimal	Percentage
$\frac{11}{100}$	0.11	11%
$1\frac{83}{100}$	1.83	183%
$\frac{5}{8}$	0.625	$62\frac{1}{2}\%$

14

- a $\frac{4}{5}$ and 0.8 b 0.475 and $\frac{19}{40}$

15

- a
i 50%, 100%, 30%
ii 0.3, $\frac{5}{10}$, 100%
- b
i 25%, 60%, 30%
ii $\frac{1}{4}$, 0.3, 60%
- c
i 37.5%, 20%, 80%
ii 20%, $\frac{3}{8}$, 0.8
- d
i 20%, 45%, 27%
ii $\frac{3}{15}$, 0.27, 45%